Homework #6 (Submit before 25-May)

❖ Ch4_No.1

The length of the laser gain medium is l, refractive index is η, the length of the laser cavity is L, and the refractive index inside the cavity expect the gain medium is η', the photon density of the gain medium is N, then proof the light with central frequency

$$\frac{dN}{dt} = \Delta n \sigma_{21} c N \frac{l}{L'} - N \frac{\delta c}{L'}$$

where $L' = \eta l + \eta'(L - l)$

❖ Ch4_No.4

For pulse Nd: YAG laser, the transmittance T_1 and T_2 of the two mirrors are 0 and 0.5 respectively. The diameter of the gain material d=0.8 cm, refractive index η=1.836, total quantum efficiency is 1, fluorescent line-width $\Delta\nu_F$=1.95×10^{11} Hz, and the spontaneous emission lifetime τ_{s2}=2.3×10^{-4} s. It is assumed that the average wavelength of the optically pumped absorption band is λ_p=0.8 μm. Estimate the threshold pump energy E_{pt} that this laser needs to absorb at the central frequency.